THE CLEBSCH CUBIC PROGRAM — IV

A binary $[78, 36, 12]$ code from the passant lines of a conic over $\mathbb{F}_{13}$

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Abstract

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2, 13)$, and form the binary incidence code on its 78 internal points using the 78 passant lines. We prove that this code has parameters $[78, 36, 12]_2$. It has exactly 364 minimum words, in four $\mathrm{PGL}(2, 13)$-orbits of size 91, and each orbit spans the code. The minimum supports reconstruct the passant incidence matrix and the six-class elliptic association scheme; their coordinate-permutation automorphism group is exactly $\mathrm{PGL}(2, 13)$. The proof combines a tangent-product exclusion in weight eight, two exhaustive but compact syndrome-disjointness checks in weight ten, the binary association algebra for spanning, and concurrence with a seven-clique census for reconstruction and four resolving anchors for symmetry.

Keywords. binary incidence code; finite conic; passant line; minimum word; association scheme; reconstruction.

1 Introduction and statement

Incidence codes often retain less geometry than the incidence structures that define them. Row operations erase the choice of parity checks, and a minimum-distance computation need not reveal why the first codewords occur. Here the minimum words recover both the parity checks and the association scheme on the coordinates. This is the fourth paper in a program on code-theoretic reconstruction around the Clebsch cubic. The first paper begins from the conic code over $\mathbb{F}_{11}$; the present $q = 13$ theorem is logically independent of the earlier papers.

Fix a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, 13)$. A line is *passant* when it is disjoint from $\mathcal{C}(\mathbb{F}_{13})$. Conic polarity identifies the 78 internal points with the 78 passant lines. Let M be their binary incidence matrix and put

$$K = \ker_{\mathbb{F}_2} M \subseteq \mathbb{F}_2^{78}.$$

The coordinate set is the set of internal points. Automorphisms below are coordinate permutations preserving the indicated code or minimum-support system.

Let

$$\mathcal{H} = \{\mathrm{supp}(c) : c \in K,\ \mathrm{wt}(c) = 12\}$$

be the minimum-support hypergraph on the unlabeled coordinate set of K.

Theorem 1.1 (Minimum-layer reconstruction)**.** *The code K has parameters $[78, 36, 12]_2$ and exactly 364 minimum words. Under $\mathrm{PGL}(2, 13)$, those words form four orbits of size*

91*: one has stabilizer isomorphic to* S_4*, and three have dihedral stabilizer of order* 24*. Every one of the four orbits spans* K*.*

From the abstract hypergraph $\mathcal{H}$*, pair and triple concurrence reconstruct the six relations of the elliptic association scheme on the coordinate set. Among the passant seven-cliques, the* 78 *with zero triple concurrence are exactly the supports of the rows of* M*. Consequently* $\mathcal{H}$ *reconstructs* M *up to independent coordinate and row permutation, and*

$$\mathrm{Aut}(K) = \mathrm{Aut}(\mathcal{H}) = \mathrm{PGL}(2, 13)$$

in its symmetric-square action on the 78 *internal points.*

Reconstruction here is a permutation statement. It does not choose an equation for the conic, label the projective points, or order the parity checks. The proof has three finite leaves: a five-row clique closure in weight eight, two syndrome-disjointness checks in weight ten, and an exact orbit/concurrence replay for the minimum layer. The reductions to those leaves and the association-algebra argument are mathematical proofs in the text.

The code belongs to the fixed-conic LDPC family introduced by Droms–Mellinger–Meyer [2]. Their general bounds, as recorded in the later comparison table of Ma–Liu–Tian [5, Table 1], specialize here to $8 \le d \le 12$. Madison and Wu proved the general nullity formula $(q-1)^2/4$ [4, Corollary 6.3]. Hollmann and Xiang constructed the elliptic association scheme afforded by the same conic action [3, Sections 3–5]. We determine the upper endpoint of the distance interval at $q = 13$, classify its minimum words, and recover the incidence and relation structures from their unlabeled support hypergraph.

2 Distance twelve

We use the model $\mathcal{C} : XZ - Y^2 = 0$. Every row and column of M has weight seven, and conic polarity identifies the two index sets. The Madison–Wu formula gives $\dim K = 36$; direct binary elimination is an independent check of rank 42.

Summing all row equations gives the all-one functional, so every word of K has even weight. If a nonzero word contains an internal point P, each of the seven passants through P contains a second support point. Those seven points are distinct. Thus every nonzero word has weight at least eight.

2.1 The tangent obstruction in weight eight

Suppose that a word has weight eight and fix a support point P. Each of the seven passants through P must contain another support point. Since only seven points remain, each passant contains exactly one of them. Applying the same argument at every support point shows that every support join is passant and that no three support points are collinear. Thus the support is an eight-arc, and every passant pencil at a support point is saturated. The seven remaining lines through each support point are therefore both the combinatorial tangents to the arc and the conic secants through that point.

For each secant line choose a nonzero linear equation. For an internal point P, put

$$T_P(X) = \prod_{\substack{\ell \ni P \\ \ell \text{ secant to } \mathcal{C}}} \ell(X).$$

For a pairwise-passant triple define

$$h(P, Q, R) = \frac{T_P(Q)T_Q(R)T_R(P)}{T_P(R)T_R(Q)T_Q(P)}.$$

Every displayed evaluation is nonzero because the three joins are passant. Rescaling a line equation multiplies one numerator and one denominator factor by the same scalar, so h is well defined. The saturated-pencil identification above makes the factors of T_P precisely the combinatorial tangents at P. Segre's lemma of tangents in the coordinate-free form of Ball–Lavrauw [1, Lemma 27] gives $h(P,Q,R) = 1$ for every triple in the hypothetical support. Indeed, its hypotheses apply to this eight-arc over the odd field $\mathbb{F}_{13}$, and with the cyclic order P, Q, R its tangent-product quotient is exactly the displayed h (not its negative; reversing the cycle only inverts it). Fix $P = (1 : 0 : 2)$. Its other seven points would form a clique in the graph on the 42 passant-join neighbors of P, where Q and R are adjacent when QR is passant and $h(P,Q,R) = 1$.

The projective map

$$\gamma(x : y : z) = (x + 6y + 9z : 7x + 9y + 3z : 10x + y + z)$$

fixes P and has order 14 on its passant-join neighbors. With indices in $\mathbf{Z}/14$, put

$$A_i = \gamma^i(1 : 1 : 7), \qquad B_i = \gamma^i(1 : 1 : 6), \qquad C_i = \gamma^i(1 : 2 : 10).$$

These three orbits are disjoint and contain all 42 neighbors. Direct substitution in the product above gives

orbit pair	$j - i \pmod{14}$
AA	$4, 6, 8, 10$
AB	$6, 7, 11, 12$
AC	$1, 3$
BB	$6, 8$
BC	$3, 5, 6, 8, 9, 11$
CC	$2, 4, 10, 12.$

An exhaustive check of the $\binom{42}{4} = 111930$ four-subsets finds exactly 70 cliques. Up to simultaneous translation of the indices, they are represented by the following five rows; the right column gives the unique common neighbor:

four-clique	unique extension
A_0, B_6, B_{12}, C_1	C_3
A_0, B_6, B_{12}, C_3	C_1
A_0, B_6, C_1, C_3	B_{12}
A_0, B_{12}, C_1, C_3	B_6
B_0, B_6, C_9, C_{11}	$A_8.$

They close to fourteen translates of one five-clique, and none extends. The local clique number is five, so a seven-clique cannot occur. This excludes weight eight.

2.2 The two weight-ten profiles

Fix one point in a hypothetical weight-ten support. Let s of the other nine points be joined to it by secants. The remaining $9 - s$ points occur in positive odd numbers in its seven passant fibres. Hence s is even and $9 - s \geq 7$, so $s = 0$ or $s = 2$. The profiles are

$$(3, 1, 1, 1, 1, 1, 1; 0), \qquad (1, 1, 1, 1, 1, 1, 1; 2),$$

where the last entry is the number of secant neighbors.

Write m_Q for the 78-bit incidence column at Q. Each of the seven passant fibres through the fixed point P contains exactly six other internal points; the other 35 internal points are its secant neighbors. For the first profile the search loops over the exceptional fibre, its three-subset, and one point from each of the other six labelled fibres. Split those six fibres, in their fixed coordinate order, into blocks of three. The two partial-XOR sets have $6^3 = 216$ and $\binom{6}{3}6^3 = 4320$ entries. Thus all $7\binom{6}{3}6^6 = 6{,}531{,}840$ labelled choices occur.

For the second profile the search chooses one point in every labelled fibre and an unordered pair among the 35 secant neighbors. Splitting off the first three fibres gives 216 distinct left syndromes and 771024 distinct right syndromes after equal XOR values are deduplicated; the latter side is generated from $6^4\binom{35}{2} = 771120$ labelled choices. Hence all $6^7\binom{35}{2} = 166{,}561{,}920$ candidates occur.

For either split, write L and R for the chosen point sets. The support has zero syndrome exactly when

$$\bigoplus_{Q\in L} m_Q = m_P \oplus \bigoplus_{Q\in R} m_Q.$$

Each side is stored as a sorted set of integer-encoded syndromes, so deduplication removes repeated encodings but no candidate syndrome. The two intersections are empty. The loops above exhaust the parity profiles, hence no candidate has zero syndrome. An independent dynamic program reconstructs the conic and incidence columns, builds full-fibre XOR sets with a different partition, and obtains the same two empty intersections.

Finally, the twelve internal points

$$(1:0:2), (1:3:2), (1:4:5), (1:1:8), (1:4:8), (1:1:7),$$
$$(1:7:12), (1:3:3), (1:9:11), (1:10:11), (1:0:5), (1:8:7)$$

have zero syndrome. Their stabilizer is dihedral of order 24. Therefore $d(K) = 12$.

3 Minimum words and the elliptic scheme

Fix one support point P. Its seven passant fibres again have six eligible points each, and it has 35 secant neighbors. Pencil parity leaves $s = 0, 2, 4$ secant neighbors. The corresponding multiplicities in the seven passant fibres, followed by the number of secant neighbors, are

$$(5, 1^6; 0), \qquad (3, 3, 1^5; 0), \qquad (3, 1^6; 2), \qquad (1^7; 4).$$

A syndrome-indexed meet in the middle exhausts four labelled domains. They loop respectively over

$$\begin{array}{c|c} (5,1^6;0) & 7\binom{6}{5}6^6 \\ (3,3,1^5;0) & \binom{7}{2}\binom{6}{3}^2 6^5 \\ (3,1^6;2) & 7\binom{6}{3}6^6\binom{35}{2} \\ (1^7;4) & 6^7\binom{35}{2}^2, \end{array}$$

where the last loop retains only disjoint secant pairs. The four splits hash, in order: three singleton fibres against the five-set and three singletons; two singletons against the two triples and three singletons; the triple and three singletons against three singletons and the secant pair; and three singletons with one secant pair against four singletons with the other pair. The key is the XOR syndrome. Every match is converted to an unordered twelve-set, checked directly against M, and inserted into a set; this deduplicates different labelled presentations without discarding a support.

The four domains contribute $0, 0, 0, 56$ distinct words. The canonical replay records four projective representatives, generates their full $\mathrm{PGL}(2, 13)$-orbits, and intersects each orbit with the supports containing P. These four fixed-point slices are disjoint, each has size

14, and their union is checked for equality with the exhaustive 56-word set. The order-28 stabilizer of the fixed point acts transitively on each slice; the stabilizer of a support inside it has order 2. Transitivity, or equivalently the count $56 \cdot 78/12 = 364$, proves global exhaustion.

The replay generates each orbit from its recorded representative and checks every support against M. The stabilizers in $\mathrm{PGL}(2, 13)$ have order 24, so each orbit has size $2184/24 = 91$. Writing r^m for m elements of order r, their profiles are

$$1^1 2^9 3^8 4^6, \qquad 1^1 2^{13} 3^2 4^2 6^2 12^4 \quad \text{(three times)}.$$

They identify one stabilizer with S_4 and the other three with dihedral groups of order 24.

For a pair or triple of coordinates, its concurrence is the number of the 364 minimum supports containing it. Pair concurrences are $7, 9, 12$ when the joining line is passant and $6, 8$ when it is secant. The full triple- concurrence histograms are distinct on the six $\mathrm{PGL}(2, 13)$-orbitals. Table 1 prints them. In the middle column, a^b means that b choices of the third point give triple concurrence a. Hence these intrinsic counts recover all six colors of the elliptic association scheme.

ρ	pair concurrence	triple-concurrence histogram	unordered pairs
1	6	$0^{26}1^{42}2^{6}3^{2}$	546
3	6	$0^{32}1^{28}2^{16}$	546
10	7	$0^{25}1^{36}2^{13}4^{2}$	546
0	8	$0^{16}1^{40}2^{20}$	273
12	9	$0^{18}1^{32}2^{24}5^{2}$	546
9	12	$0^{7}1^{34}2^{27}3^{4}5^{4}$	546

Table 1: The six pair orbitals recovered from the minimum supports.

The three colors $\rho = 9, 10, 12$, equivalently pair concurrences $12, 7, 9$, have passant joins; their union recovers the passant graph. A complete maximal-clique enumeration in this union finds 1716 cliques of size seven. Exactly 78 have zero concurrence on all 35 of their triples. Direct incidence comparison identifies them with the supports of the 78 rows of M. Pair and triple concurrence are intrinsic to the minimum-support hypergraph, so this reconstruction is independent of the original projective labels.

4 Spanning and automorphisms

For internal points $P = (x_P : y_P : z_P)$ and Q, set

$$\Delta(P) = y_P^2 - x_P z_P, \qquad \beta(P, Q) = 2y_P y_Q - x_P z_Q - z_P x_Q,$$

and

$$\rho(P, Q) = \frac{\beta(P, Q)^2}{\Delta(P)\Delta(Q)}.$$

On distinct internal points, ρ takes the six values $0, 1, 3, 9, 10, 12$. Its level relations are the elliptic scheme; write A_r for their adjacency matrices. Polarity identifies A_0 with M, up to row order.

The needed part of the integral intersection table reduces modulo two to

$$\begin{aligned} A_0^2 &= I + A_9 + A_{10} + A_{12}, \\ A_0 A_r &= 0 \quad (r = 9, 10, 12), \\ A_9^2 &= A_{10}, \qquad A_{10}^2 = A_{12}, \qquad A_{12}^2 = A_9. \end{aligned} \tag{1}$$

These are parity reductions of intersection numbers. The $\mathrm{PGL}(2,13)$-action is transitive on each relation, so fixing one point and substituting one representative of every relation checks all entries.

Put $B = A_9$. Since $A_0 B = 0$, the image of B lies in K. If $x \in K$ and $Bx = 0$, then (1) gives

$$0 = A_0^2 x = (I + B + B^2 + B^4)x = x.$$

Thus B is injective on the 36-dimensional space K, so $\operatorname{im} B = K$, and B, B^2, B^4 all have rank 36. If N_i is the 91-by-78 support matrix of the i-th minimum-word orbit, direct pair counting in one representative gives

$$(N_1^{\mathsf{T}} N_1, N_2^{\mathsf{T}} N_2, N_3^{\mathsf{T}} N_3, N_4^{\mathsf{T}} N_4) = (A_9, A_9, A_{12}, A_{10}).$$

Every N_i therefore has rank at least 36. Its rows lie in K, so each orbit spans K.

It remains to determine the scheme automorphisms. Use the four anchors

$$P_0 = (1:0:2), \quad P_1 = (1:1:7), \quad P_2 = (1:0:7), \quad P_3 = (1:1:3).$$

The rigidity mechanism is summarized by

$$\underbrace{(P_0, P_1, P_2)}_{\text{(10,3,9) regular orbit}} \quad \Longrightarrow \quad \underbrace{P_3}_{\text{(3,1,9) is unique}} \quad \Longrightarrow \quad \underbrace{P \longmapsto \big([P = P_i], \rho(P, P_i)\big)_{i=0}^{3}}_{\text{injective on the 78 points}}.$$

The first three satisfy

$$\big(\rho(P_0, P_1), \rho(P_0, P_2), \rho(P_1, P_2)\big) = (10, 3, 9).$$

There are $78 \cdot 14 \cdot 2 = 2184$ ordered triples with this pattern: A_{10} has valency 14 and $p^{10}_{3,9} = 2$. Substitution in the symmetric-square action shows that the stabilizer of (P_0, P_1, P_2) in $\mathrm{PGL}(2,13)$ is trivial. Since the group also has order 2184, it acts simply transitively on these triples.

For a fixed triple, P_3 is the unique point with relation signature $(3,1,9)$. The four values

$$\big(\rho(P, P_0), \rho(P, P_1), \rho(P, P_2), \rho(P, P_3)\big),$$

with equality to an anchor recorded on the diagonal, distinguish all 78 points. Consequently a scheme automorphism can be composed with a unique element of $\mathrm{PGL}(2,13)$ to fix the first three anchors; it then fixes the fourth and every point. Thus the reconstructed scheme has automorphism group $\mathrm{PGL}(2,13)$.

Every coordinate automorphism of K preserves $\mathcal{H}$. Conversely, $\mathcal{H}$ spans K, so every automorphism of $\mathcal{H}$ preserves K. The two coordinate-permutation groups coincide. This completes the proof of Theorem 1.1.

5 Proof and verification boundary

Parity and pencil geometry give the low-weight reductions; the tangent product controls weight eight; the association algebra explains spanning; and concurrence with four anchors gives reconstruction and rigidity. Three steps remain finite: the two weight-ten syndrome exclusions, the fixed-point classification of the minimum words, and the bounded clique and anchor checks.

The accompanying `verification/` directory contains the canonical weight-ten generator, an independent replay with a separately reconstructed incidence matrix, and the main exact replay. The last records the four minimum-word representatives and checks the clique closure, rank, orbit exhaustion, concurrence reconstruction, spanning identities, and anchors. The entry point `verify_evidence.py` checks the recorded byte counts and SHA-256 hashes before running all three programs. Every computation is deterministic and uses exact finite-field or binary arithmetic. From the paper root, the public command

```
make check
```

ends with `Paper-IV q=13 evidence: PASS` and then builds the warning-free PDF. The persistent repository-relative references are `verification/README.md` for the exact programs and `lean-certificates/README.md` for the formal package. The Python evidence uses only the standard library. On the release x86-64 Linux workstation, Python 3.13.12 runs `make evidence` in about 6.2 seconds with peak resident memory below 100 MiB; the full command additionally uses the Nix-provided TeX Live environment to build the PDF.

The shared Lean library formalizes the normalized conic and incidence code, the two parity-profile reductions, the tangent graph, the binary association argument, and the reconstruction and anchor interfaces. The paper-owned package checks the finite leaves. Table 2 records the division of proof.

claim	proof mode	exact boundary
dimension 36	published theorem; Lean theorem	Madison–Wu gives the formula; exact elimination and semantic recovery maps prove rank 42 at $q = 13$.
weights $8, 10$	human reduction; classical input; trusted finite execution	Pencil parity and Segre's lemma give the finite graph and two syndrome domains; Lean native evaluation and independent Python check those leaves.
minimum layer	Lean native evaluation; trusted Python; human transport	Both executions exhaust the 56 fixed-point words; projective transitivity and the double count give all 364.
orbit spanning	human proof; Lean	The identities in (1) and the four checked Gram matrices prove that each orbit spans $\ker A_0 = K$.
reconstruction and symmetry	trusted finite execution; human transport; Lean theorem	Concurrence recovers the relation colors and row family; the four-anchor theorem classifies scheme automorphisms. Concurrence equivariance carries support automorphisms to the scheme.

Table 2: Proof modes and trust boundaries for Theorem 1.1.

The paper-owned aggregate namespace is `PassantCodeQ13.Gates.Main`; its terminals are

```
incidenceRankAndCodeDimension
fixedPointWeightTwelveExhaustion
minimumOrbitsSpanRhoZeroKernel
recoveredRowFamilyIsUnique
ellipticSchemeAutomorphismsAreProjective.
```

The shared low-weight namespace is `RelativeConicArcs.Gates.PassantCodeQ13`; its terminals are

```
weightEight_semantic_transport
arbitrary_weightTen_profile_transport.
```

Native evaluation compiles and runs the finite decision procedures rather than reducing them entirely in the kernel. Under Lean 4.32.0-rc1, the tracked `#print axioms` audit therefore exposes each such use as a declaration-local axiom; the symbolic transport theorems remain kernel-checked relative to those finite leaves. Thus "Lean theorem" in Table 2 means kernel-checked relative to its printed axioms, whereas “Lean native evaluation” and “trusted Python” are executions. Hash checks identify the executed sources and outputs but do not turn either execution into a kernel-checked certificate; no finite leaf in the table is described that way. The formal scope stops at two human transports. Projective transitivity and the double count carry the fixed-point exhaustion to all 364 minimum words. Concurrence carries minimum-support automorphisms to automorphisms of the polar-relation scheme. Segre’s lemma of tangents is the cited classical input to the formal weight-eight transport.

6 Conclusion

The code determines a row space, but its minimum words recover more: they distinguish the original parity-check rows inside that space. At $q = 13$, concurrence recovers the elliptic relations, the binary association algebra explains why every minimum orbit spans, and four anchors force the full projective symmetry.

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